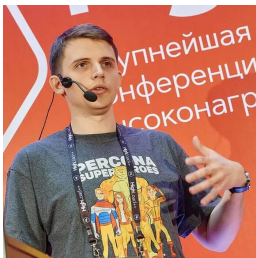


MySQL Up and Running in 30 minutes

Sergey Kuzmichev
Vinicius Grippa



Who we are



Sergey Kuzmichev

Databases, performance, reliability. Infrastructure and Performance Architect at [Investing.com](https://investing.com). We [are hiring](#).

Linkedin: <https://www.linkedin.com/in/skuzmichev>

Telegram: @arronax

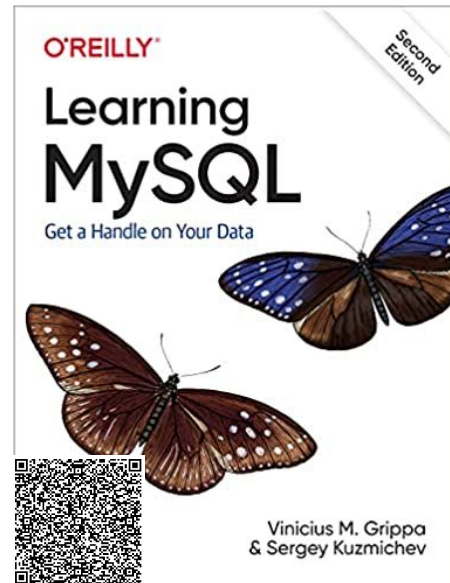
GitHub: <https://github.com/arronax>



Vinicius Grippa

Senior Database Engineer at [Percona](https://percona.com).

Linkedin: <https://www.linkedin.com/in/vinicius-grippa/>



Agenda

- What is a relational database management system (RDBMS)? How different it is from a NoSQL database?
- Is SQL useful for Data Science?
- Deploying a MySQL instance
- MySQL GUI tools
- Dataset examples

What is a relational database management system (RDBMS)?

- Data is organized in tables, where columns represent attributes and rows represent records.
- Relation in RDBMS is a table, not the FK or another constraint.
- Schema is strict and omnipresent. Each table is defined to hold specific data.
- Data is traditionally accessed mainly via the SQL.
- SQL stands for Structured Query Language, a language used to interact with a relational database. It can be used to read and write (create, update, delete) data in a Relational Database Management System (RDBMS).

What is a relational database management system (RDBMS)?

The diagram illustrates a table structure with the following annotations:

- Column (attribute):** Points to the header row of the table.
- Table (relation):** Points to the entire table structure.
- Row (tuple):** Points to one of the data rows.
- Primary key:** Points to the **CustomerID** column.
- Data value:** Points to the value **Green** in the **LastName** column of the last row.

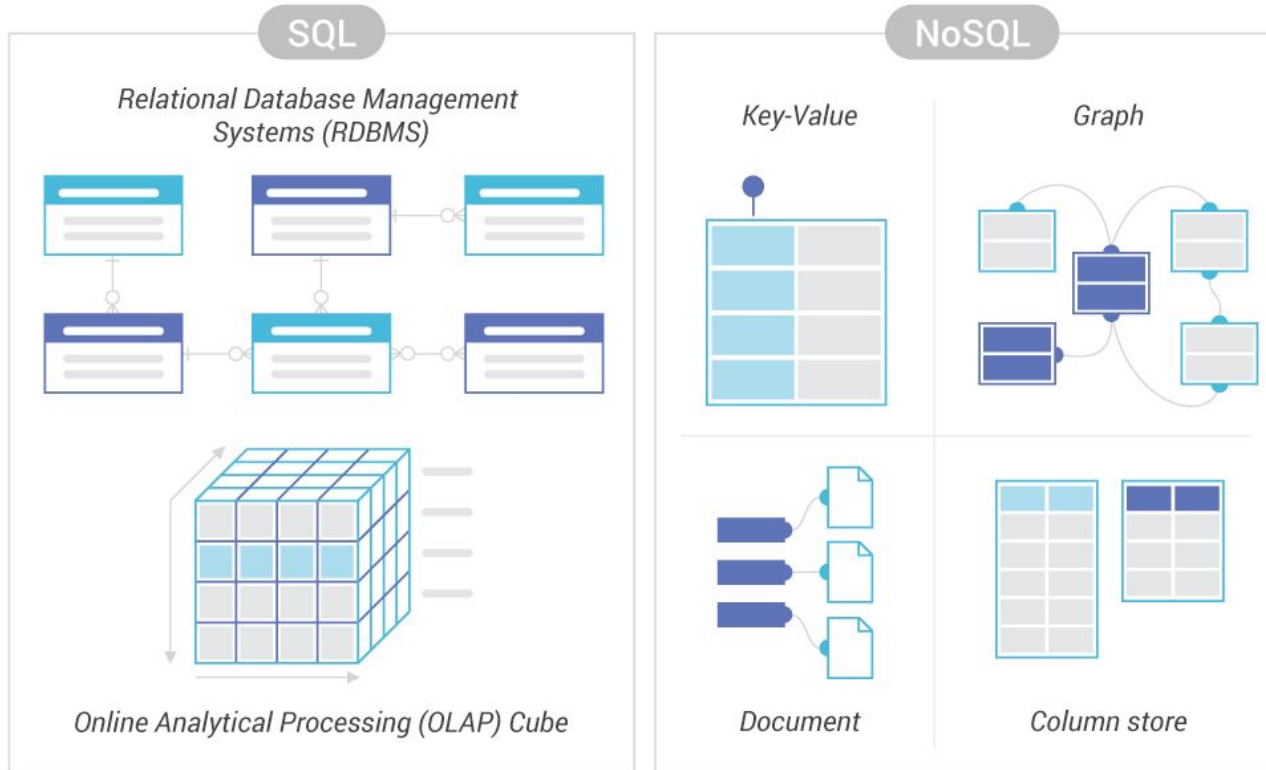
CustomerID	FirstName	LastName	Birthdate
XY001	John	Doe	April 18, 1929
BR092	Mary	Green	March 4, 1980
PD500	Francesca	de la Gillebert	September 12, 1959
WI308	John	Green	March 4, 1980

Source: <https://medium.com/@oliverknocklein/visualizing-sql-a-beginners-guide-to-relational-databases-c2dcfda79ea4>

How different it is from a NoSQL database?

- While not a mandatory trait, most NoSQL databases will have no schema or a relaxed schema.
- Constraints and table dependencies come naturally in an RDBMS, but they do not define the RDBMS.
- NoSQL can store different types of data.

How different it is from a NoSQL database?

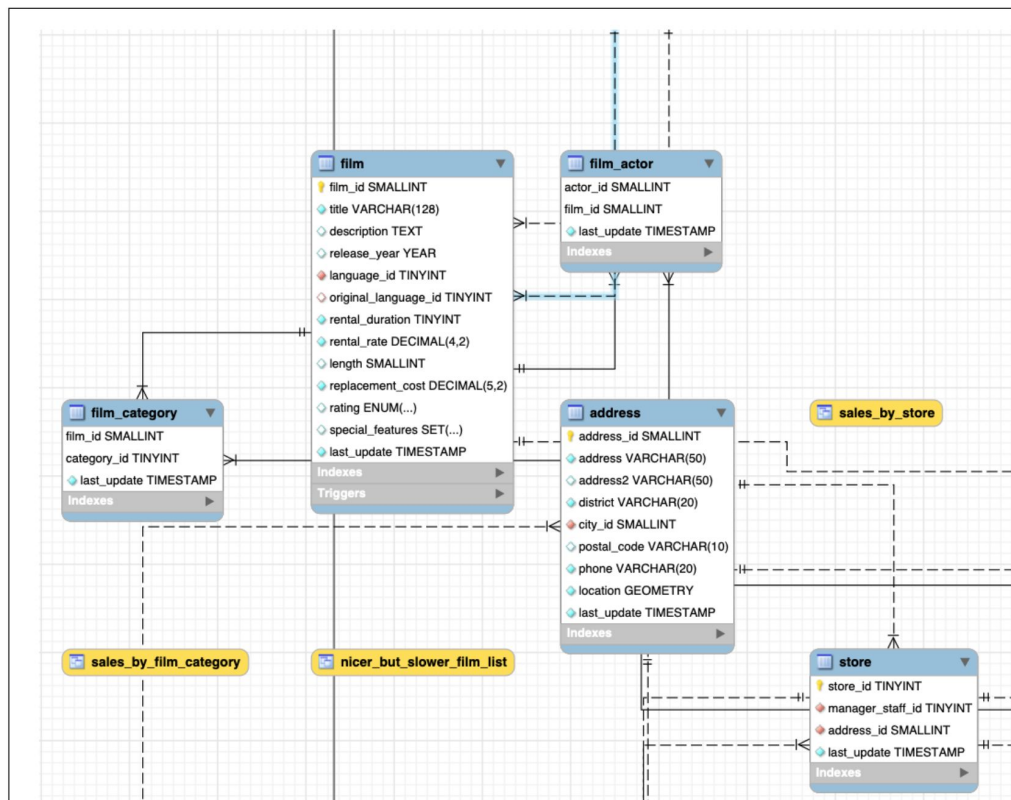


Source: <https://www.scylladb.com/learn/nosql/nosql-vs-sql/>

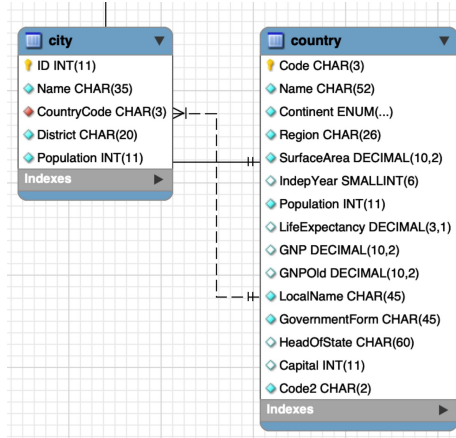
Is SQL useful for Data Science?

- SQL remains the ideal choice for many CRM, business intelligence tools, finance and in office operations.
- A Data Scientist needs SQL in order to handle structured data (extract, manipulate and transform). This structured data is stored in relational databases.

Is SQL useful for Data Science?



Structured Query Language (SQL)



```
SELECT Name FROM country;
```

```
SELECT Name FROM city;
```

```
SELECT city.Name, country.Name FROM country INNER JOIN city ON  
CountryCode=Code;
```

Deploying a MySQL instance



Deploying a MySQL instance

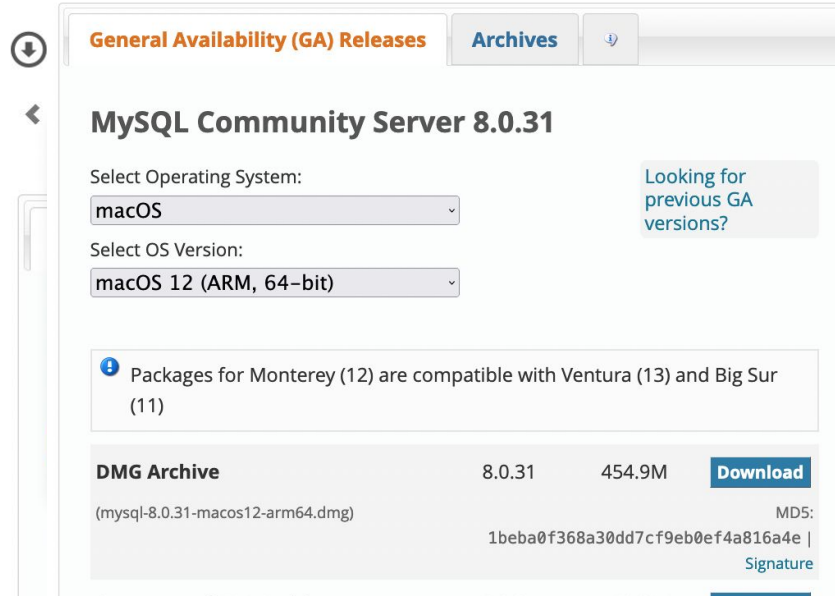
- There are many ways to deploy or install a database.
- It is possible to install in a server, virtualized instances (EC2, Compute Engine, Virtualbox).
- Or more recently ways as docker, LXC, Kubernetes, podman.
- In both options, you can automate the deployment (Terraform, Ansible, Vagrant, ...)
- And more recently, DBaaS (database-as-a-service).

Deploying a MySQL instance

DEMO

Deploying MySQL locally

- [MySQL Community Server installers or packages are available for most OSes](#)
- MacOS and Windows have graphical installers

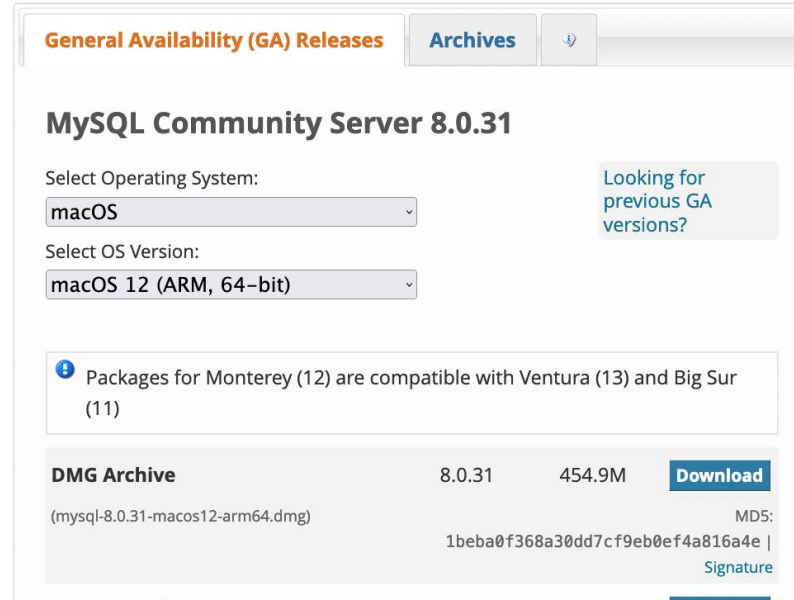


Deploying MySQL locally – MacOS

- Windows installation is similar to one outlined here
- [Dedicated installer for Windows](#)

Deploying MySQL locally – macOS

- Pick macOS as the OS
- Pick your CPU (Intel is x86, Apple Silicon is ARM)
- Get DMG



The screenshot shows the MySQL Community Server 8.0.31 download page. At the top, there are tabs for 'General Availability (GA) Releases' (selected), 'Archives', and a help icon. Below the tabs, the title 'MySQL Community Server 8.0.31' is displayed. Underneath, there are two dropdown menus: 'Select Operating System:' with 'macOS' selected, and 'Select OS Version:' with 'macOS 12 (ARM, 64-bit)' selected. To the right of these dropdowns is a link that says 'Looking for previous GA versions?'. Below the dropdowns, there is a message box with an information icon stating: 'Packages for Monterey (12) are compatible with Ventura (13) and Big Sur (11)'. At the bottom, there is a table with one row showing the 'DMG Archive' for version 8.0.31, which is 454.9M in size. The filename is '(mysql-8.0.31-macos12-arm64.dmg)'. To the right of the filename, there is a 'Download' button. Below the filename, there is an MD5 hash: '1beba0f368a30dd7cf9eb0ef4a816a4e |'. To the right of the MD5 hash, there is a 'Signature' link.

General Availability (GA) Releases Archives ⓘ

MySQL Community Server 8.0.31

Select Operating System:
macOS ▾

Select OS Version:
macOS 12 (ARM, 64-bit) ▾

Looking for previous GA versions?

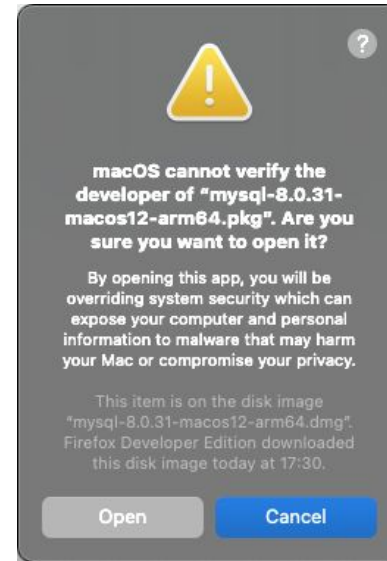
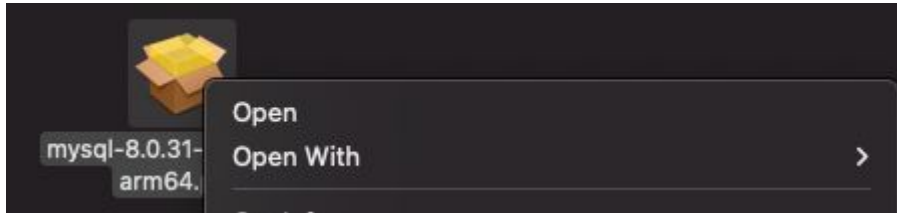
ⓘ Packages for Monterey (12) are compatible with Ventura (13) and Big Sur (11)

DMG Archive (mysql-8.0.31-macos12-arm64.dmg)	8.0.31	454.9M	Download
--	--------	--------	--------------------------

MD5:
1beba0f368a30dd7cf9eb0ef4a816a4e |
[Signature](#)

Deploying MySQL locally – MacOS

- Bypass developer verification using forbidden knowledge
- Right mouse click (double-click on touchpad, or ctrl+click a single mouse button)



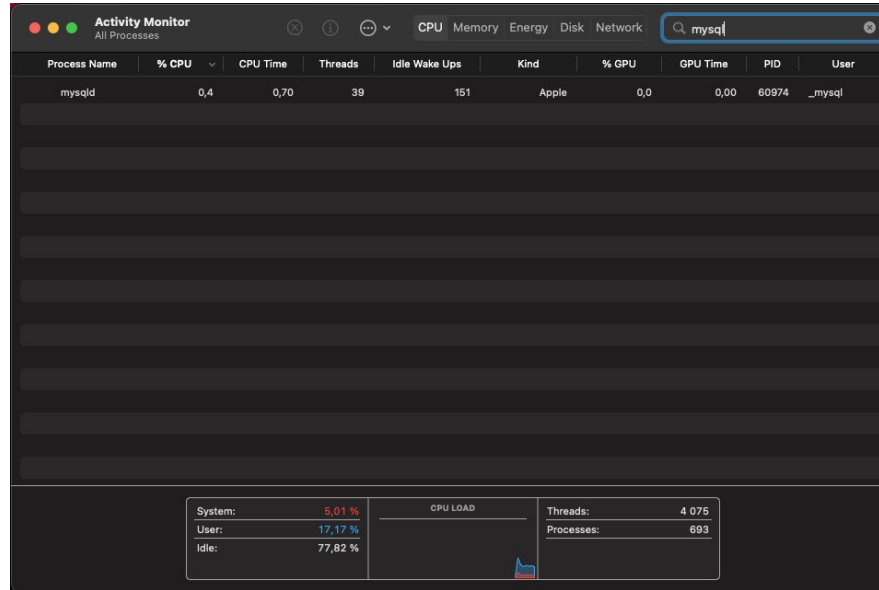
Deploying MySQL locally – MacOS

- Press next and continue everywhere, it's that simple
- We recommend using Legacy Password (only for local envs, not prod!)



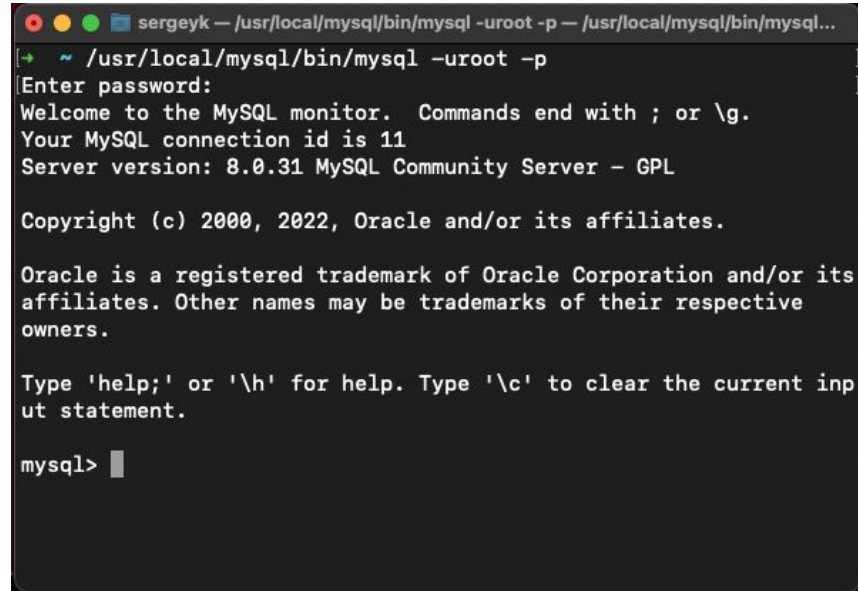
Deploying MySQL locally – MacOS

- `mysqld` process – MySQL Daemon – is running



Deploying MySQL locally – MacOS

- MySQL CLI can be used to verify the installation

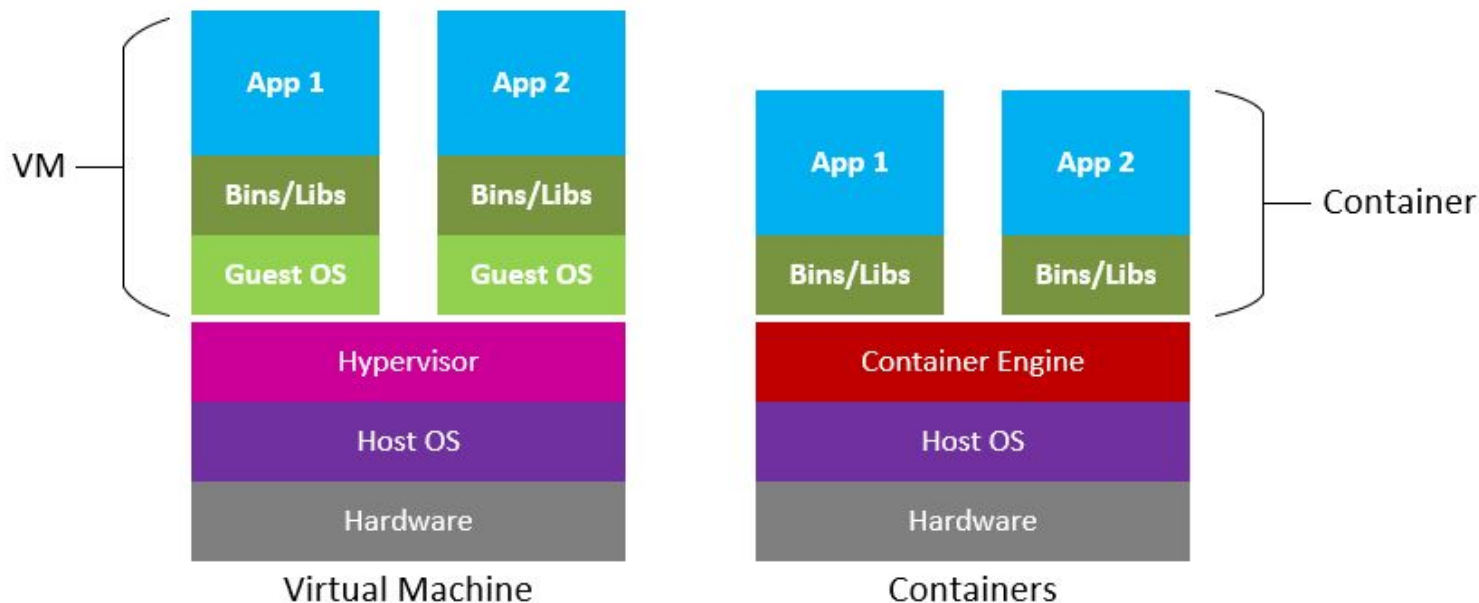
A terminal window with a dark background and light text. The window title bar shows a red, yellow, and green window control icon, followed by the text 'sergeyk — /usr/local/mysql/bin/mysql -uroot -p — /usr/local/mysql/bin/mysql...'. The terminal content shows the command '/usr/local/mysql/bin/mysql -uroot -p' being executed. It prompts for a password, then displays a welcome message, connection ID, server version (8.0.31 MySQL Community Server - GPL), copyright information, and usage instructions. The prompt 'mysql>' is visible at the bottom with a cursor.

```
sergeyk — /usr/local/mysql/bin/mysql -uroot -p — /usr/local/mysql/bin/mysql...  
➔ ~ /usr/local/mysql/bin/mysql -uroot -p  
[Enter password:  
Welcome to the MySQL monitor.  Commands end with ; or \g.  
Your MySQL connection id is 11  
Server version: 8.0.31 MySQL Community Server - GPL  
  
Copyright (c) 2000, 2022, Oracle and/or its affiliates.  
  
Oracle is a registered trademark of Oracle Corporation and/or its  
affiliates. Other names may be trademarks of their respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current inp  
ut statement.  
  
mysql> █
```



Deploying MySQL with Docker

- Docker uses OS-level virtualization to deliver software in containers. It is possible to have many containers running in the same host. It is usually faster to deploy than a VM.



Deploying MySQL with Docker

DEMO

Deploying MySQL with Docker



Docker Desktop

Install Docker Desktop – the fastest way to containerize applications.

Download Docker Desktop

 Intel Chip



Apple Chip



Windows



Linux

Deploying MySQL with Docker



Docker Desktop Update to latest Sign in

Containers [Give feedback](#)

A container packages up code and its dependencies so the application runs quickly and reliably from one computing environment to another. [Learn more](#)

☐ Only show running containers

		NAME	IMAGE	STATUS	PORT(S)	STARTED	ACTIONS
☐		ps bcef0ef37fc0	percona/percona-server:8.0	Running		1 day ago	☐ ⋮ 
☐		mysql-latest 799345691bb7	mysql/mysql-server:latest	Running	6033:3306	1 day ago	☐ ⋮ 



Deploying MySQL with Docker

```
$ docker run --name mysql-latest -p 6033:3306 -e  
MYSQL_ROOT_PASSWORD='learning_mysql' -d mysql/mysql-server:latest  
  
$ docker exec -ti mysql-latest mysql -uroot -plearning_mysql
```

Deploying MySQL with Docker



```
vinnie-macpro:Downloads vgrippa$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
38ad18246717	mysql/mysql-server	"/entrypoint.sh mysql..."	About a minute ago	Up About a minute (healthy)	33060-33061/tcp, 0.0.0.0:6033->3306/tcp	mysql-latest

Deploying Percona Server/MariaDB with Docker



```
$ docker run -d --name ps -e MYSQL_ROOT_PASSWORD=root percona/percona-server:8.0
```

```
$ docker run --name mariadb -p 3307:3306 -e MYSQL_ROOT_PASSWORD=password -d mariadb
```

Deploying MySQL with Docker



CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
afff5c823258	mariadb	"docker-entrypoint.s..."	4 seconds ago	Up 2 seconds	0.0.0.0:3307->3306/tcp	mariadb
740fc048d19f	percona/percona-server:8.0	"/docker-entrypoint...."	2 minutes ago	Up 2 minutes	3306/tcp, 33060/tcp	ps
38ad18246717	mysql/mysql-server	"/entrypoint.sh mysql..."	7 minutes ago	Up 7 minutes (healthy)	33060-33061/tcp, 0.0.0.0:6033->3306/tcp	mysql-latest

Interfacing with MySQL – CLI

- Convenient on a server
- Probably not the best way to learn

Interfacing with MySQL – CLI

DEMO

Interfacing with MySQL – MySQL Workbench

- Similar to the MySQL Server itself, Oracle provides [Workbench installers](#)
- x86 app works on ARM Macs



General Availability (GA) Releases Archives Info

MySQL Workbench 8.0.31

Select Operating System:

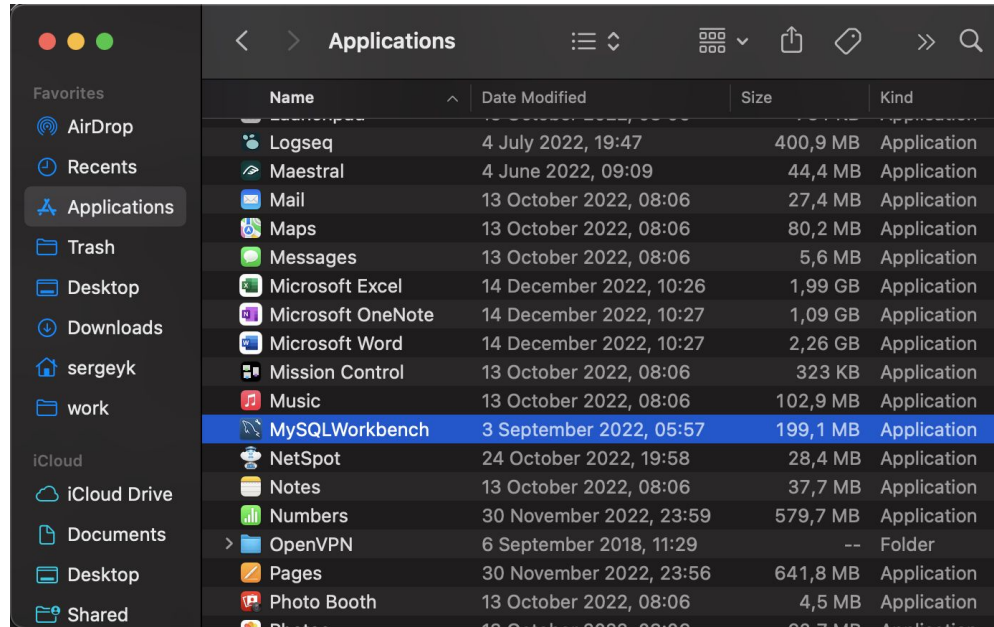
macOS

 Packages require Big Sur (11.1 or newer)

macOS (x86, 64-bit), DMG Archive	8.0.31	113.0M	Download
(mysql-workbench-community-8.0.31-macos-x86_64.dmg)	MD5: 57927c4341d3ae5addb1ad82ac9647e3 Signature		

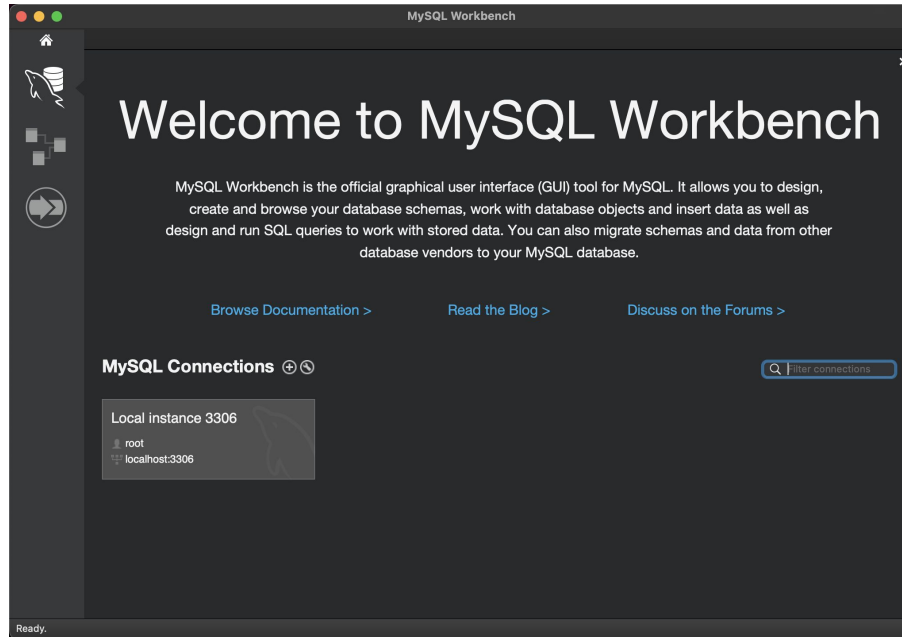
Interfacing with MySQL – MySQL Workbench

- After following the standard dmg installation, find Workbench in Apps list



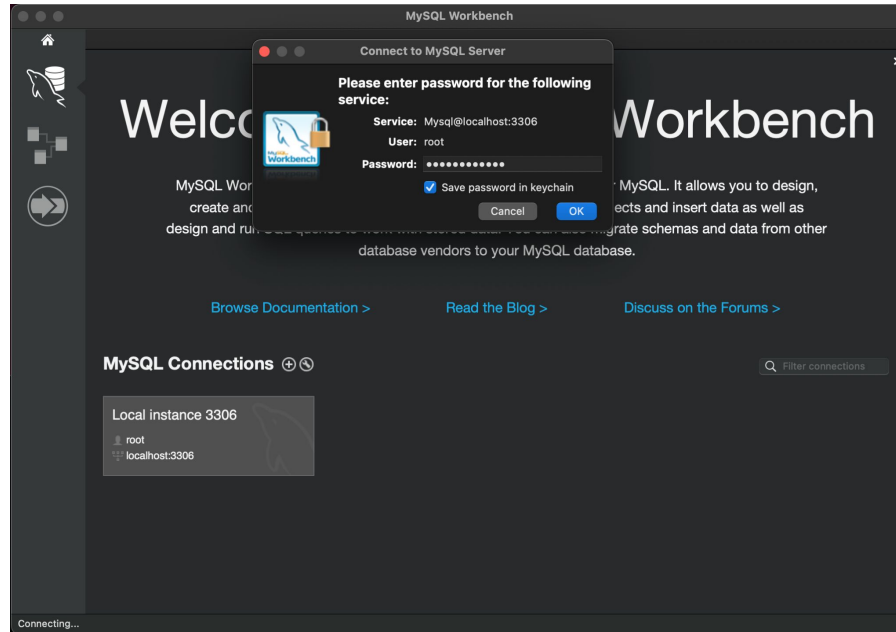
Interfacing with MySQL – MySQL Workbench

- Default Workbench startup screen has a Local connection



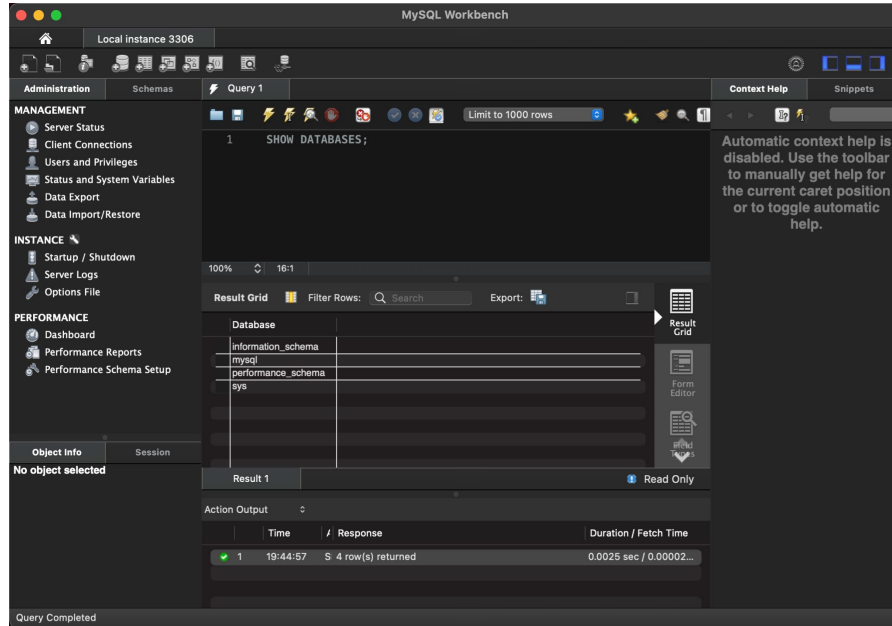
Interfacing with MySQL – MySQL Workbench

- Hopefully, you remember the password you set when installing MySQL



Interfacing with MySQL – MySQL Workbench

- By default, a fresh MySQL installation has no data



Demo Datasets and where to find them

- By default, a fresh MySQL installation has no data
- There are many datasets available online
- Here are [some from Oracle itself](#)
- We will show Sakila and Employee databases

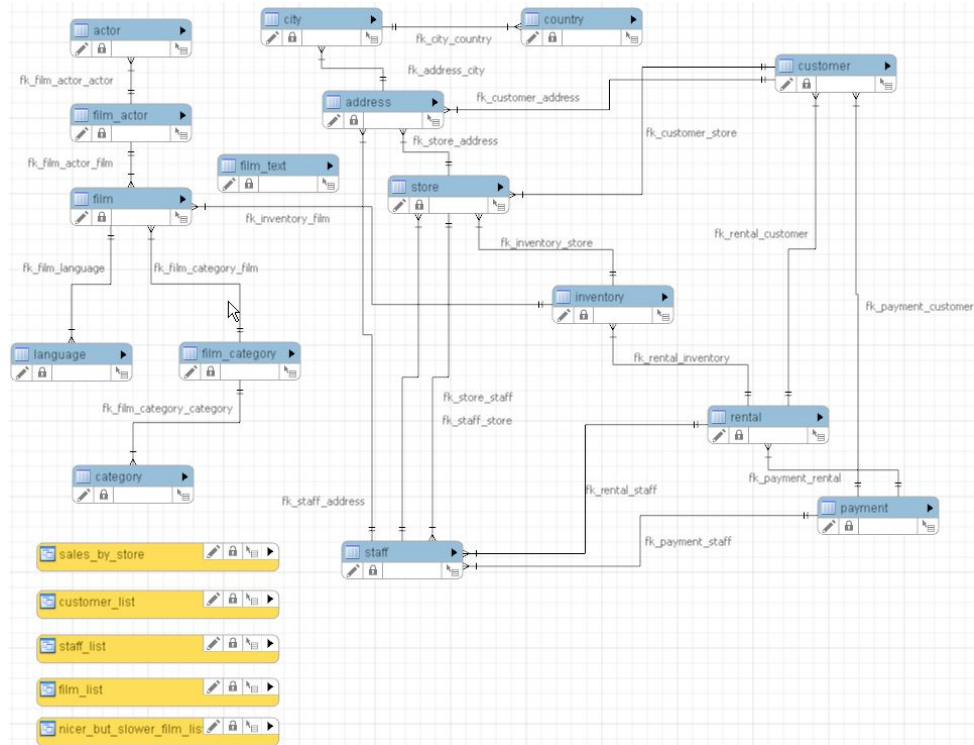
Example Databases

Title	DB Download	HTML Setup Guide	PDF Setup Guide
employee data (large dataset, includes data and test/verification suite)	GitHub	View	US Ltr A4
world database	TGZ Zip	View	US Ltr A4
world_x database	TGZ Zip	View	US Ltr A4
sakila database	TGZ Zip	View	US Ltr A4
airportdb database (large dataset, intended for MySQL on OCI and HeatWave)	TGZ Zip	View	US Ltr A4
menagerie database	TGZ Zip		

Sakila database

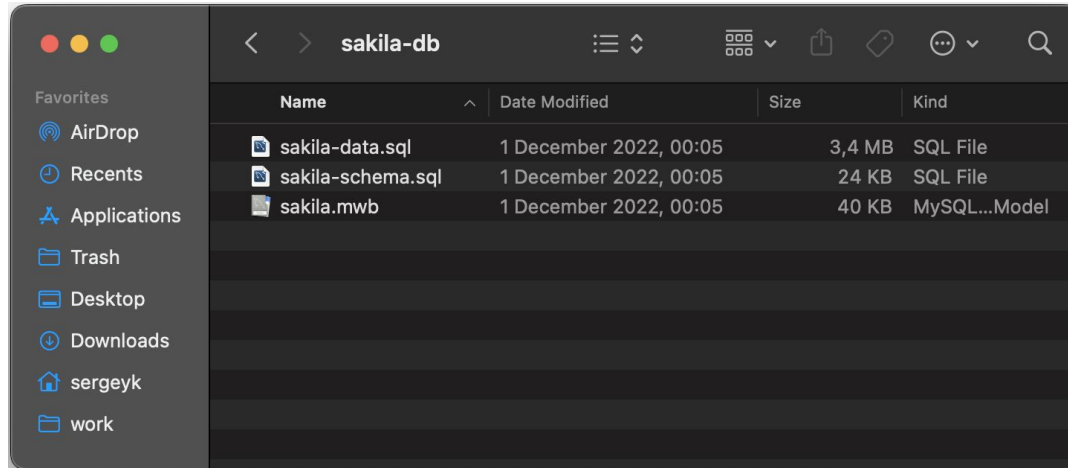
- Sakila is a classic example database
- Simple and small, yet allows for some complicated queries
- Rental company data

Sakila database



Sakila database – installation

- [Download from mysql.com](#)
- Unpack (both macOS and Windows support zip files by default)
- Navigate to the unpacked directory

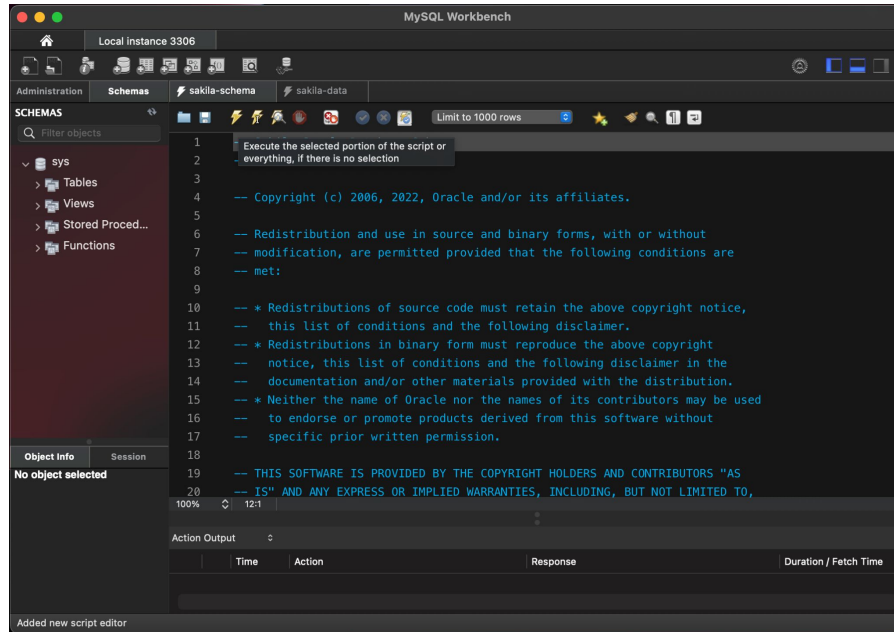


Sakila database – installation

- We will use MySQL Workbench to install the data
- Open two scripts
- Either from Workbench
- Or just doubleclick from the OS (Explorer or Finder)

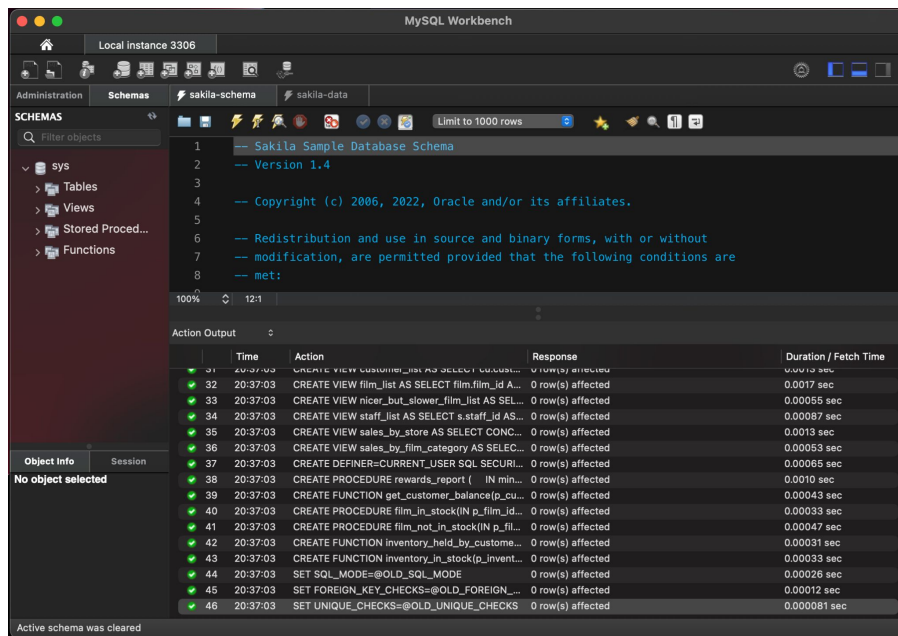
Sakila database – installation

- Execute sakila-schema



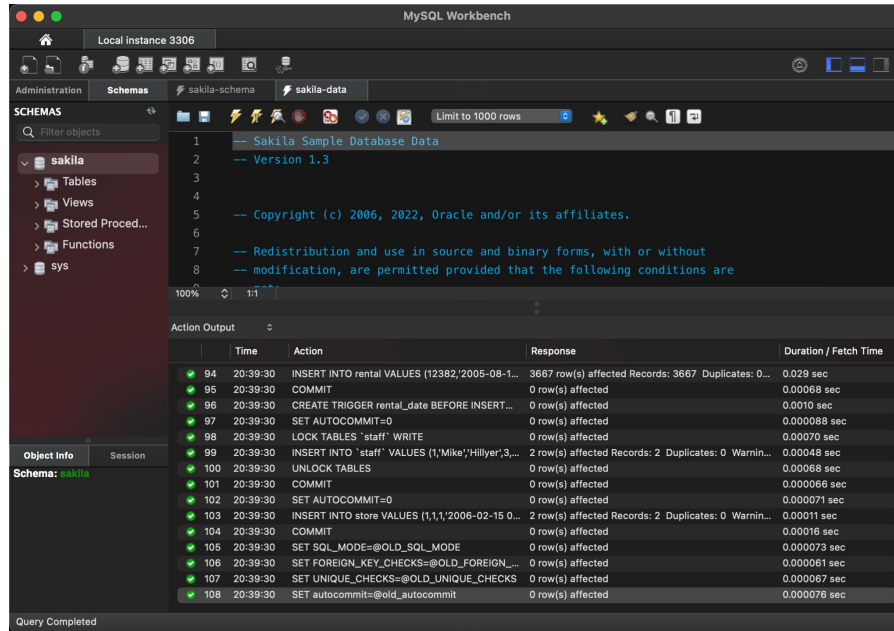
Sakila database – installation

- All action output items (but one warning) should be green



Sakila database – installation

- Execute sakila-data and observe the output



The screenshot shows the MySQL Workbench interface with the 'sakila-data' script being executed. The 'Schemas' pane on the left shows the 'sakila' database selected. The main editor displays the script content, and the 'Action Output' pane at the bottom shows the execution results.

Script Content:

```
1 -- Sakila Sample Database Data
2 -- Version 1.3
3
4
5 -- Copyright (c) 2006, 2022, Oracle and/or its affiliates.
6
7 -- Redistribution and use in source and binary forms, with or without
8 -- modification, are permitted provided that the following conditions are
```

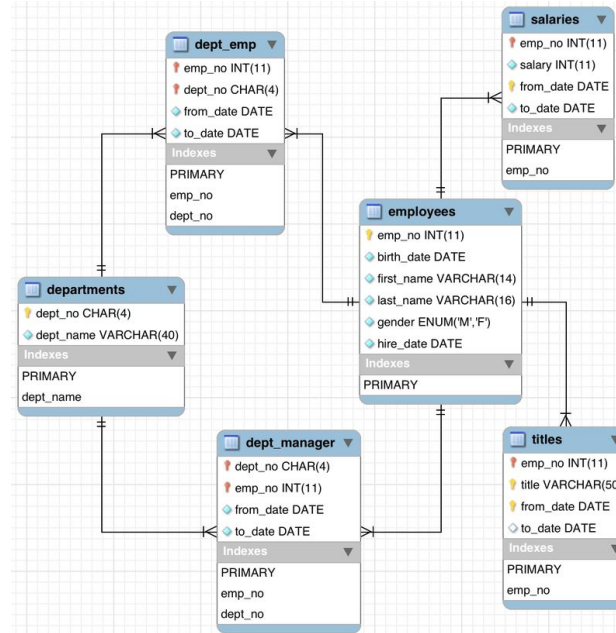
Action Output:

	Time	Action	Response	Duration / Fetch Time
94	20:39:30	INSERT INTO rental VALUES (12382,'2005-08-1...	3667 row(s) affected Records: 3667 Duplicates: 0...	0.029 sec
95	20:39:30	COMMIT	0 row(s) affected	0.00068 sec
96	20:39:30	CREATE TRIGGER rental_date BEFORE INSERT...	0 row(s) affected	0.0010 sec
97	20:39:30	SET AUTOCOMMIT=0	0 row(s) affected	0.000088 sec
98	20:39:30	LOCK TABLES 'staff' WRITE	0 row(s) affected	0.00070 sec
99	20:39:30	INSERT INTO 'staff' VALUES (1,'Mike','Hillyer',3...	2 row(s) affected Records: 2 Duplicates: 0 Warnin...	0.00048 sec
100	20:39:30	UNLOCK TABLES	0 row(s) affected	0.00068 sec
101	20:39:30	COMMIT	0 row(s) affected	0.000068 sec
102	20:39:30	SET AUTOCOMMIT=0	0 row(s) affected	0.000071 sec
103	20:39:30	INSERT INTO store VALUES (1,1,'2006-02-15 0...	2 row(s) affected Records: 2 Duplicates: 0 Warnin...	0.00011 sec
104	20:39:30	COMMIT	0 row(s) affected	0.00016 sec
105	20:39:30	SET SQL_MODE=@OLD_SQL_MODE	0 row(s) affected	0.000073 sec
106	20:39:30	SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN...	0 row(s) affected	0.000061 sec
107	20:39:30	SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS	0 row(s) affected	0.000067 sec
108	20:39:30	SET autocommit=@old_autocommit	0 row(s) affected	0.000076 sec

Query Completed

Employee database

- Another great sample database
- Larger size than Sakila, though still on a small side (~150MiB)
- Simpler schema structure

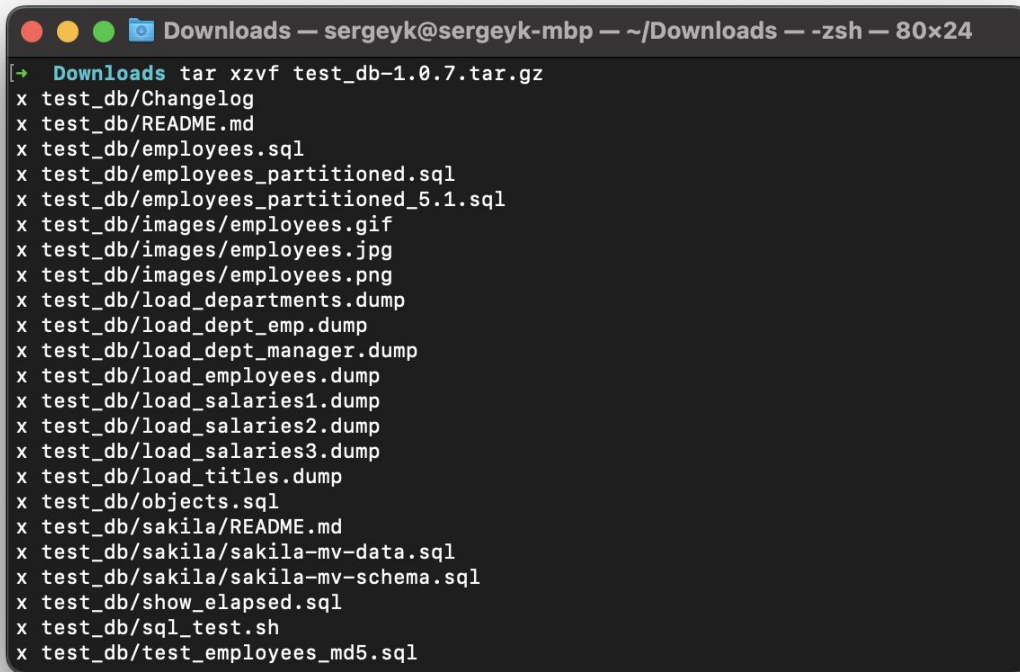


Employee database – installation

- We will need to use a terminal and MySQL CLI
- Download Employee database [from github](#)
- Package is a tar.gz file, not native for Windows or macOS
- Both OSes (Win from version 10) support tar, however
- Command for Windows and macOS is going to be the same

Employee database – installation

- `tar xzvf test_db-1.0.7.tar.gz`

A terminal window titled "Downloads — sergeyk@sergeyk-mbp — ~/Downloads — -zsh — 80x24". The prompt is "[→ Downloads]". The command "tar xzvf test_db-1.0.7.tar.gz" has been executed, resulting in a list of files being extracted, each preceded by an "x".

```
[→ Downloads] tar xzvf test_db-1.0.7.tar.gz
x test_db/Changelog
x test_db/README.md
x test_db/employees.sql
x test_db/employees_partitioned.sql
x test_db/employees_partitioned_5.1.sql
x test_db/images/employees.gif
x test_db/images/employees.jpg
x test_db/images/employees.png
x test_db/load_departments.dump
x test_db/load_dept_emp.dump
x test_db/load_dept_manager.dump
x test_db/load_employees.dump
x test_db/load_salaries1.dump
x test_db/load_salaries2.dump
x test_db/load_salaries3.dump
x test_db/load_titles.dump
x test_db/objects.sql
x test_db/sakila/README.md
x test_db/sakila/sakila-mv-data.sql
x test_db/sakila/sakila-mv-schema.sql
x test_db/show_elapsed.sql
x test_db/sql_test.sh
x test_db/test_employees_md5.sql
```

Employee database – installation

- `mysql -u root -p < employees.sql`

```
test_db — sergeyk@sergeyk-mbp — ../loads/test_db — zsh — 80x24
[→ Downloads cd test_db ]
[→ test_db /usr/local/mysql/bin/mysql -uroot -p < employees.sql ]
[Enter password: ]
INFO
CREATING DATABASE STRUCTURE
INFO
storage engine: InnoDB
INFO
LOADING departments
INFO
LOADING employees
INFO
LOADING dept_emp
%INFO
LOADING dept_manager
INFO
LOADING titles
INFO
LOADING salaries
data_load_time_diff
00:00:19
→ test_db %
```

Interfacing with MySQL – MySQL Workbench

- Now that we have sample data
- Let's explore and run some queries
- It's super easy

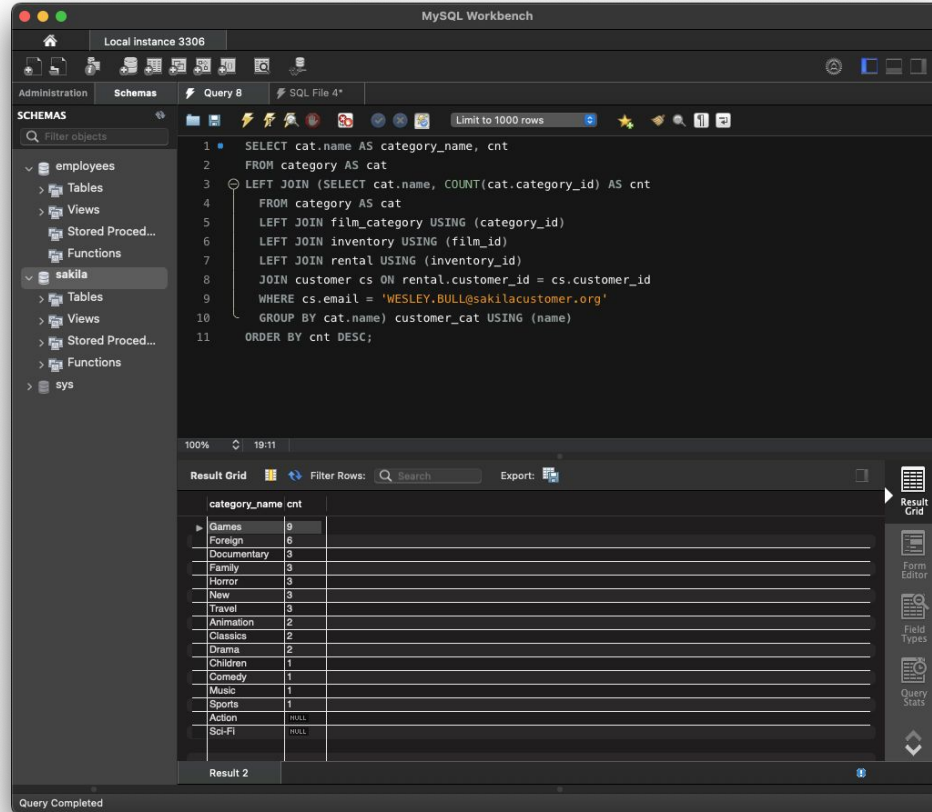
Interfacing with MySQL – Sakila sample query

- Even though Sakila is simple, it allows for fairly complicated queries
- Break down rented films by category for a specific customer
- Include 0 or NULL where they rented no films
- All basic SQL but might not be as straightforward to write initially

Interfacing with MySQL – Sakila sample query

```
SELECT cat.name AS category_name, cnt
FROM category AS cat
LEFT JOIN (SELECT cat.name, COUNT(cat.category_id) AS cnt
FROM category AS cat
LEFT JOIN film_category USING (category_id)
LEFT JOIN inventory USING (film_id)
LEFT JOIN rental USING (inventory_id)
JOIN customer cs ON rental.customer_id = cs.customer_id
WHERE cs.email = 'WESLEY.BULL@sakilacustomer.org'
GROUP BY cat.name) customer_cat USING (name)
ORDER BY cnt DESC;
```

Interfacing with MySQL – Sakila sample query



The screenshot displays the MySQL Workbench interface for a local instance 3306. The left sidebar shows the 'SCHEMAS' tree with the 'sakila' database selected. The main editor window shows a SQL query (Query 8) that joins the 'category', 'film_category', 'inventory', 'rental', and 'customer' tables to find the number of rentals per category, ordered by count in descending order. The query is as follows:

```
1 SELECT cat.name AS category_name, cnt
2 FROM category AS cat
3 LEFT JOIN (SELECT cat.name, COUNT(cat.category_id) AS cnt
4 FROM category AS cat
5 LEFT JOIN film_category USING (category_id)
6 LEFT JOIN inventory USING (film_id)
7 LEFT JOIN rental USING (inventory_id)
8 JOIN customer cs ON rental.customer_id = cs.customer_id
9 WHERE cs.email = 'WESLEY.BULL@sakilacustomer.org')
10 GROUP BY cat.name) customer_cat USING (name)
11 ORDER BY cnt DESC;
```

The 'Result Grid' at the bottom shows the results of the query, with columns 'category_name' and 'cnt'. The results are as follows:

category_name	cnt
Games	9
Foreign	6
Documentary	3
Family	3
Horror	3
New	3
Travel	3
Animation	2
Classics	2
Drama	2
Children	1
Comedy	1
Muscle	1
Sports	1
Action	HULL
Sci-Fi	HULL

The status bar at the bottom indicates 'Query Completed'.

Interfacing with MySQL – Employees sample query

- Employees allows playing around with more complicated queries
- Still really small and simple
- Let's try some window functions
- And CTEs
- Show minimum and maximum salaries per department with each employee
- Find percentile bucket where an employee's salary falls

Interfacing with MySQL – Employees sample query

- Employees allows playing around with more complicated queries
- Still really small and simple
- Let's try some window functions
- And CTEs

Interfacing with MySQL – Employees sample query

- Employees allows playing around with more complicated queries
- Still really small and simple
- Let's try some window functions
- And CTEs
- Rank salaries per department and show alongside min and avg salaries

Interfacing with MySQL – Employees sample query

```
WITH current_salaries AS (  
    SELECT emp_no, salary  
    FROM salaries  
    WHERE to_date = '9999-01-01'  
),  
current_titles AS (  
    SELECT emp_no, title  
    FROM titles  
    WHERE to_date = '9999-01-01'  
),  
employees_dept_sal AS (  
    SELECT emp.first_name, emp.last_name, dep.dept_name, ct.title, cs.salary  
    FROM employees emp  
        JOIN current_salaries cs USING (emp_no)  
        JOIN current_titles ct USING (emp_no)  
        JOIN dept_emp de USING (emp_no)  
        JOIN departments dep USING (dept_no)  
    )  
SELECT  
    first_name, last_name, dept_name, title, salary,  
    ROUND((percent_rank() OVER (PARTITION BY dept_name ORDER BY salary ASC)*100),2) salary_pct,  
    MIN(salary) OVER(partition by dept_name) AS min_salary,  
    AVG(salary) OVER(partition by dept_name) AS avg_salary  
FROM  
    employees_dept_sal  
ORDER BY salary DESC, salary_pct DESC
```

Interfacing with MySQL – Employees sample query

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'SCHEMAS' tree with 'employees' expanded. The central pane shows 'Query 2' with the following SQL code:

```
17 JOIN departments dep USING (dept_no)
18 )
19 SELECT
20     first_name, last_name, dept_name, title, salary,
21     ROUND((percent_rank() OVER (PARTITION BY dept_name ORDER BY salary ASC)+100),2) salary_pct,
22     MIN(salary) OVER(partition by dept_name) AS min_salary,
23     AVG(salary) OVER(partition by dept_name) AS avg_salary
24 FROM
25     employees_dept_sal
26 ORDER BY salary DESC, salary_pct DESC
```

The bottom pane shows the 'Result Grid' with 20 rows of data. The columns are: first_name, last_name, dept_name, title, salary, salary_pct, min_salary, and avg_salary. The data is sorted by salary in descending order.

first_name	last_name	dept_name	title	salary	salary_pct	min_salary	avg_salary
Tokuyasu	Pesch	Sales	Senior Staff	158220	100	39426	88842.1590
Honesty	Mukaidonono	Sales	Senior Staff	156286	100	39426	88842.1590
Xiahua	Whitcomb	Sales	Senior Staff	155709	100	39426	88842.1590
Sanjai	Luders	Sales	Senior Staff	155513	99.99	39426	88842.1590
Tsutomu	Alameldin	Sales	Senior Staff	155190	99.99	39426	88842.1590
Willard	Baca	Sales	Senior Staff	154459	99.99	39426	88842.1590
Lidong	Meriste	Sales	Senior Staff	154376	99.99	39426	88842.1590
Charmane	Griswold	Sales	Senior Staff	153715	99.98	39426	88842.1590
Weijjing	Chenoweth	Sales	Senior Staff	152710	99.98	39426	88842.1590
Weicheng	Hatcliff	Sales	Senior Staff	152687	99.98	39426	88842.1590
Shin	Birdsall	Sales	Senior Staff	152412	99.98	39426	88842.1590
Mitsuyuki	Stanfel	Sales	Senior Staff	152220	99.97	39426	88842.1590
Mohamm...	Moehrke	Sales	Senior Staff	150740	99.97	39426	88842.1590
Ibibia	Junet	Sales	Senior Staff	150345	99.97	39426	88842.1590
Lansino	Kambil	Sales	Senior Staff	150052	99.97	39426	88842.1590

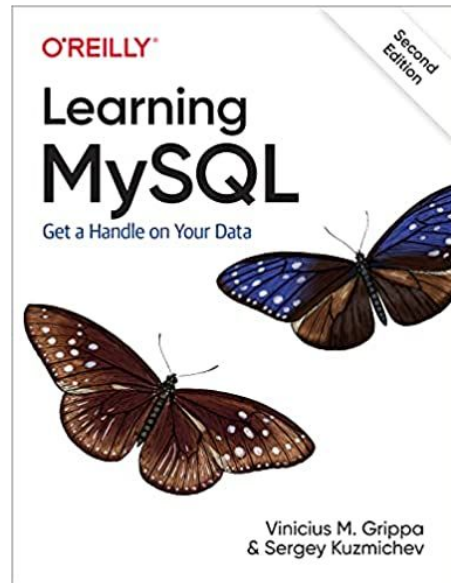
The status bar at the bottom indicates 'Query Completed' and 'Read Only'.

QUESTIONS?



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Thank you

Grazie

Obrigado

Gracias